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PROJECT : Fleet Dispatch Tracker

1. Technical Environment

- **Language: PHP (Procedural/Lightweight OOP)**
- **Database: MongoDB (NoSQL)**
- **Testing Tool: Postman (for API verification)**
- **Package Manager: Composer (for MongoDB driver)**

2. Objectives/Features:

1. **DB Connections & Console-Based: Connects directly to MongoDB; core execution is driven by terminal commands (php cli.php).**
2. **No Frontend: Zero HTML, CSS, or JavaScript. Strictly a backend architecture.**
3. **API Checking: Includes an api.php router to process HTTP requests, outputting JSON for Postman verification.**
4. **Operations Point (Create): Implements Create/Insert functions for the 4 core entities: Drivers, Vehicles, Shipments, and Warehouses.**
5. **Full CRUD: Supports all required operations: Drop (Delete), Update, Insert, and Read across the database.**
6. **Insert Functionality: Direct database insertion methods are mapped to both CLI inputs and API POST requests.**
7. **Database Mappings (Relationships): Successfully implements all four relationship types using NoSQL document structures rather than SQL joins.**

3. NoSQL Database Schema & Mappings

Instead of traditional relational tables, the project uses MongoDB's document architecture to prove an understanding of NoSQL database design:

- **One-to-One (1:1): *Embedded Data.* A Vehicle document embeds its specific Registration_Details (capacity, fuel type) directly inside it.**
- **One-to-Many (1:N): *Parent Referencing.* A Warehouse document holds an array of vehicle_ids for all trucks currently parked there.**
- **Many-to-One (N:1): *Child Referencing.* Multiple individual Shipment documents reference a single warehouse_id where they are stored.**
- **Many-to-Many (M:N): *Junction Logging.* A Dispatch_Logs collection captures the exact driver_id, vehicle_id, and timestamp, mapping the reality that drivers operate multiple trucks and trucks are driven by multiple drivers over time.**

4. Execution (How it Runs)

The system uses a shared database logic file (db.php) that is triggered in two different ways depending on the user's need:

- **Action: Dispatching a Truck (Let)**

  Via Console (Internal Use):

- ***Input:*** **php cli.php dispatch "DRV_101" "TRK_55" "Warehouse_North"**
- ***Output:*** **Terminal prints a success message and updates the database.**

  Via Postman (External API Use):

- ***Input:*** **POST request to http://localhost/api.php?action=dispatch with JSON payload.**
    - ***Output:*** **Postman receives a formatted {"status": "success"} JSON response, and the database is updated.**